Advanced Javascript frameworks(Angular)

TASK MANAGEMENT

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ABSTRACT:

Task Management Tracker is a web-based task management application developed using Angular and TypeScript. The application is designed to help users efficiently organize, track, and manage their daily tasks through a structured and interactive interface. It provides a systematic way to handle tasks by allowing users to create, update, prioritize, and monitor task progress.

The project focuses on implementing essential task management features such as task creation, categorization, priority assignment, due date tracking, and completion status management. Angular's component-based architecture is used to ensure modularity, scalability, and easy maintenance of the application. The use of Angular Material enhances the visual appeal and consistency of the user interface.

This project demonstrates the practical application of modern frontend technologies in building a real-world productivity tool. The Task Management Tracker can be further enhanced by integrating backend services, user authentication, notifications, and advanced analytics, making it suitable for real-time task and project management.

OBJECTIVES:

- To design and develop a web-based task management application using Angular.
- To provide a simple, interactive, and user-friendly interface for managing tasks.
- To implement core task operations such as creation, updating, and deletion of tasks.
- To support task prioritization, categorization, and due date management.
- To ensure responsive design for smooth usage across different devices
- To apply Angular's component-based architecture and TypeScript for structured development.

SCOPE OF THE PROJECT:

- Development of a frontend-based task management system.
- Implementation of CRUD (Create, Read, Update, Delete) operations for tasks.
- Support for task categorization, priority levels, and completion status
- Design of a responsive and visually consistent user interface
- Storage of task data using browser local storage.

- Scope for future enhancements such as backend integration, authentication, and notifications.

TOOLS & TECHNOLOGIES USED:

Tools/Technologies	Purpose
Angular (v20)	Frontend framework
TypeScript	Application logic
Angular Material	UI components and styling
JSON Server	Mock backend
RxJS	Asynchronous data handling
HTML5	Page structure
CSS3	Styling and layout
Node.js	Runtime environment
Angular CLI	Project creation and management
VS Code	Code editor
Chrome DevTools	Testing and debugging

APPLICATION STRUCTURE OVERVIEW:

- The application is developed using Angular with a component-based architecture.
- The src folder contains all core application files.
- Components handle the user interface and music playback features.
- Services manage data handling and business logic.
- Routing is used for navigation between different views.
- Modules are used to organize and manage application features.
- The structure supports scalability and easy future enhancements.

MODULE DESCRIPTION:

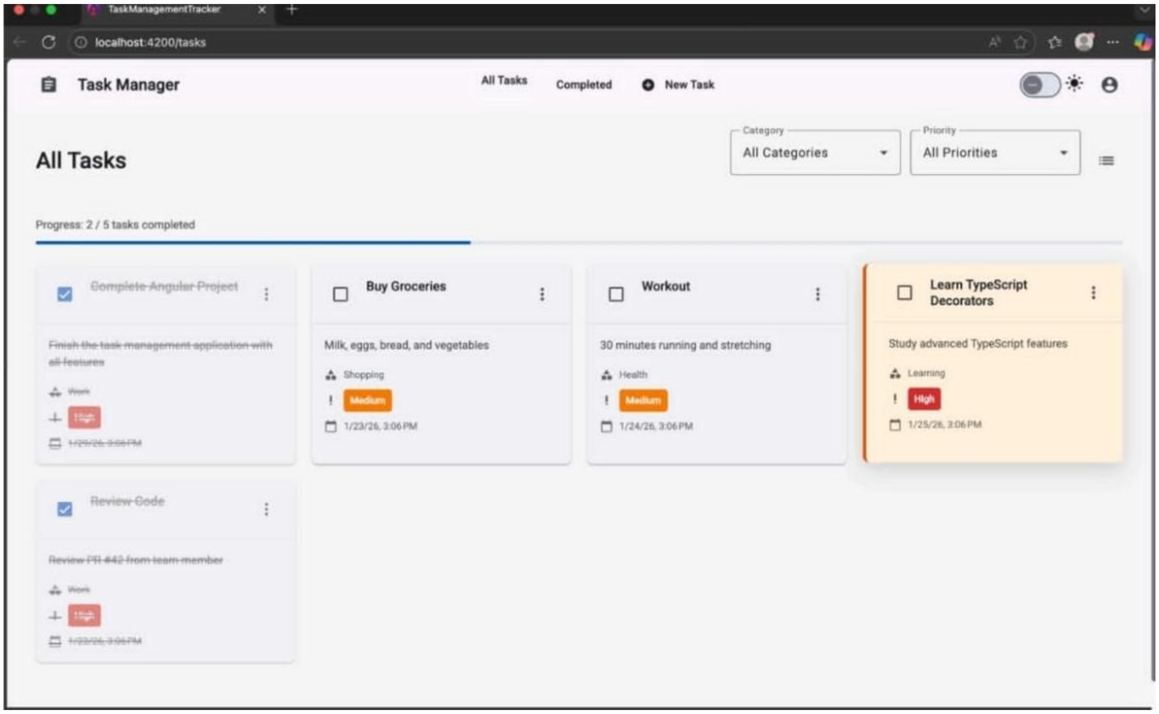
1)Task List & Filter Module

Module Description:
The Task List & Filter Module is responsible for displaying all tasks created by the user. It presents tasks in a card-based layout and allows users to monitor task progress easily. This module acts as the central view where users can manage and track all active tasks.

KEY FUNCTIONS:

- Displays all pending and active tasks
- Shows task details such as title, description, category, priority, and due date
- Provides filtering options based on category and priority
- Allows users to mark tasks as completed
- Displays task completion progress

SCREENSHOT:



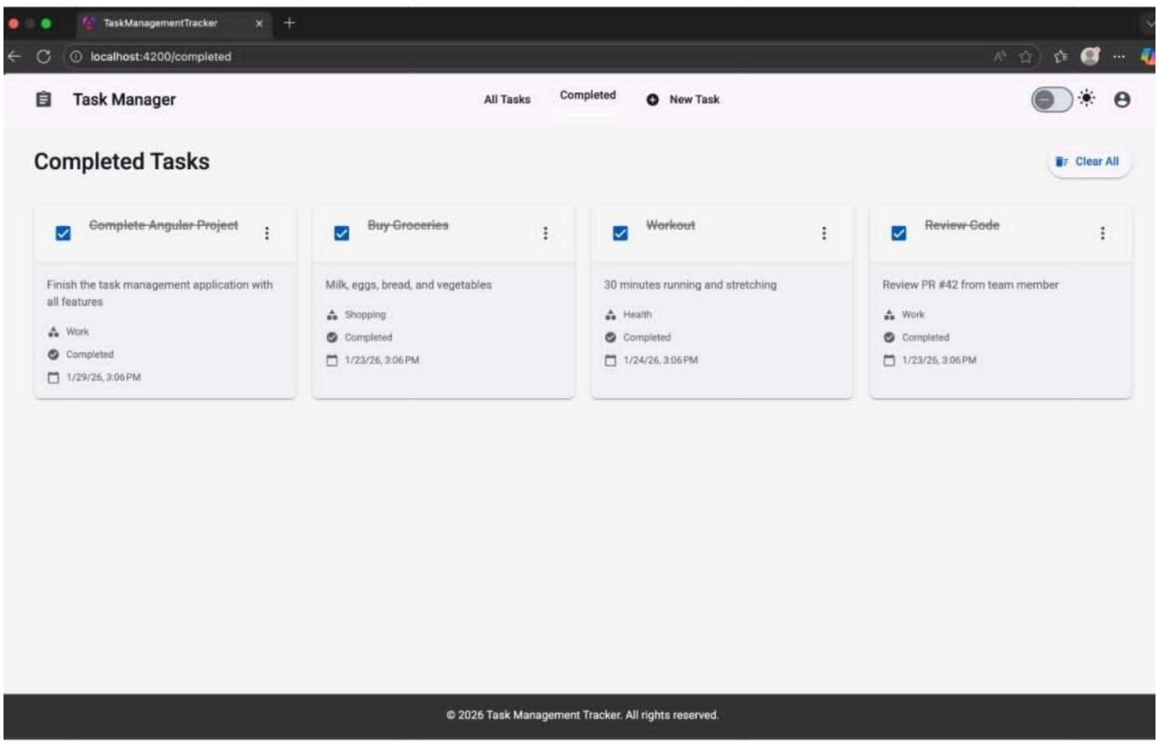
2) Completed Tasks Module

Module Description:
The Completed Tasks Module manages and displays tasks that have been marked as completed. It helps users review finished tasks and maintain a clear separation between pending and completed work.

KEY FUNCTIONS :

- Displays all completed tasks
- Shows completion status and task details
- Allows users to clear completed tasks
- Helps track productivity and task completion history

SCREENSHOT:



3) Task Management Module (Add Task Module)

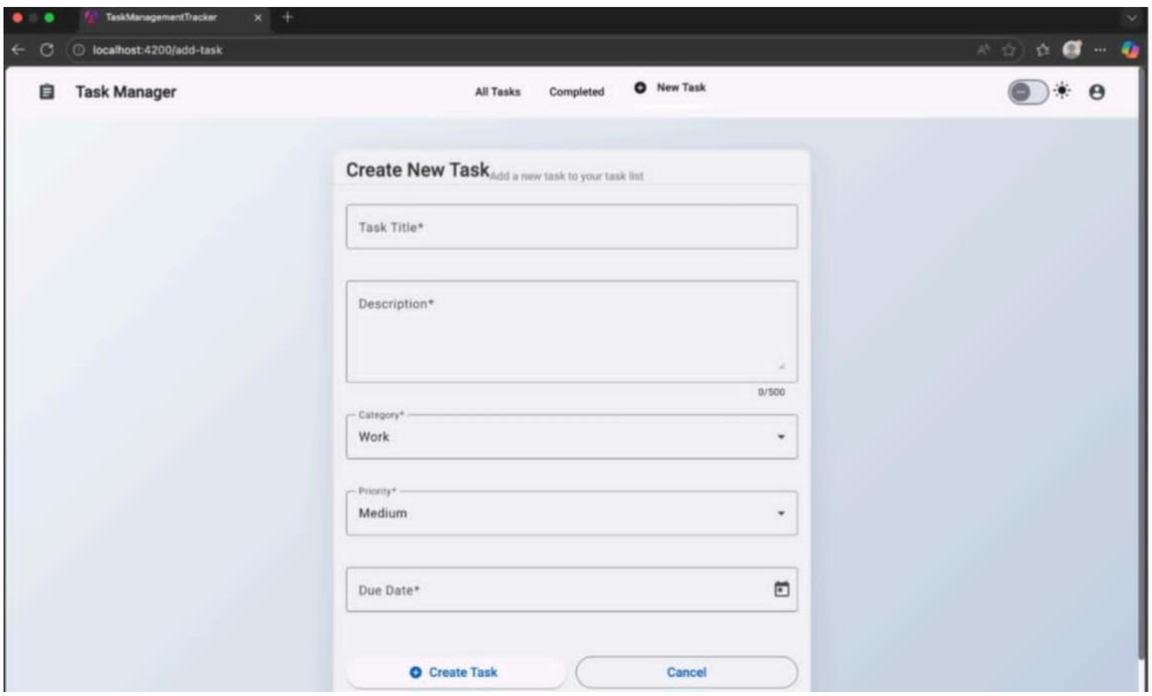
Module Description:

The Task Management Module handles the creation and modification of tasks. It provides a structured form where users can enter task details and add them to the task list.

KEY FUNCTIONS:

- Allows users to create new tasks
- Accepts task title and description
- Enables category and priority selection
- Allows setting a due date
- Validates input before task creation

SCREENSHOT:



ANGULAR & TYPESCRIPT OVERVIEW:

Angular is a powerful frontend framework used to build scalable single-page applications using components, services, and routing. It enables structured development and efficient state management. TypeScript, a strongly typed superset of JavaScript, enhances code quality by providing type safety, improved readability, and easier debugging. Together, Angular and TypeScript enable the development of a maintainable and high-performance task management application.

KEY FEATURES:

- Create, edit, and delete tasks
- Assign priority levels (High, Medium, Low)
- Categorize tasks for better organization
- Set and manage due dates
- Mark tasks as completed or pending
- Filter and sort tasks based on priority and status
- Light and dark theme support
- Responsive design for desktop and mobile devices

CHALLENGES FACED & SOLUTIONS:

CHALLENGES	SOLUTION
Component communication	Used Angular services for data sharing
Asynchronous data handling	Implemented RxJS Observables
UI consistency	Adopted Angular Material components
Routing between pages	Implemented Angular Routing Module
State management for playback	Managed state using shared services
Responsive layout	Used CSS Flexbox and Grid

OUTCOME:

⇨ Successfully developed a functional task management web application.
⇨ Implemented task CRUD operations with filtering and sorting features.
⇨ Achieved modular and scalable application architecture.
⇨ Gained practical experience in Angular, TypeScript, and frontend development.

FUTURE ENHANCEMENTS:

- User authentication and role-based access
- Backend and database integration
- Task reminders and notifications
- Advanced analytics and task reports
- Cloud synchronization and multi-device support

CONCLUSION:

The Task Management Tracker project demonstrates the effective use of Angular and TypeScript in building a real-world productivity application. The system provides an organized and efficient approach to managing tasks and improves user productivity. With further enhancements such as backend integration and notifications, the application can be extended into a complete task and project management solution.

REFERENCES:

- ⇨ Angular Official Documentation – https://angular.io/docs
- ⇨ TypeScript Official Documentation – https://www.typescriptlang.org/docs
- ⇨ Angular Material Documentation – https://material.angular.io
- ⇨ RxJS Documentation – https://rxjs.dev/guide/overview
- ⇨ Node.js Official Website – https://nodejs.org
- ⇨ JSON Server Documentation – https://github.com/typicode/json-server
- ⇨ MDN Web Docs – HTML5 & CSS3 – https://developer.mozilla.org
- ⇨ Visual Studio Code Documentation – https://code.visualstudio.com/docs